

A note on investment costs and the value of capital goods

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1 Model

It is possible to derive from the standard RBC model the endogenous asset price denoted $Q_t = \lambda_t^k / \lambda_t^c$, defined as the ratio between the marginal cost of raising new capitals (Lagrangian multiplier on capital accumulation constraint λ_t^c) and the marginal utility of consumption (Lagrangian multiplier on budget constraint λ_t^c). Variable Q_t also refers to the well-known Tobin's Q published in Tobin (1969). Tobin's Q is the ratio between a physical asset's market value and its replacement value, this ratio has considerable macroeconomic implications as it constitutes the standard bridge between financial markets and markets for goods and services. In steady state, the ratio must be normalised to one to be consistent with asset pricing theory, such that $\bar{Q} = 1$.

To create a wedge between the price of goods and the price of capital goods, we introduce a congestion cost on the accumulation process of investment goods as in Christiano and al. (2005):

$$K_{jt} = \varepsilon_t^I I_{jt} (1 - AC_{jt}^I) + (1 - \delta) K_{jt-1} \quad (1)$$

where AC_{jt}^I is an investment cost function such that:

$$AC_{jt}^I = \frac{\chi^I}{2} \left(\frac{I_{jt}}{I_{jt-1}} - 1 \right)^2 \quad (2)$$

$$= f \left(\frac{I_{jt}}{I_{jt-1}} \right) \quad (3)$$

This cost function has no implications on the steady state of the model, with $\overline{AC}^I = 0$.

1.1 Households

The Lagrangian problem is given by:

$$\begin{aligned} \mathcal{L}_t = & \mathbb{E}_t \sum_{\tau=0}^{\infty} \varepsilon_{t+\tau}^C \beta^\tau \left\{ \frac{C_{jt+\tau}^{1-\sigma_C}}{1-\sigma_C} - \chi \frac{H_{jt+\tau}^{1+\sigma_L}}{1+\sigma_L} \right. \\ & + \lambda_{t+\tau}^c [W_{t+\tau} H_{jt+\tau} + Z_{t+\tau} K_{jt-1+\tau} + R_{t-1+\tau} B_{jt-1+\tau} - C_{jt+\tau} - I_{jt+\tau} - B_{jt+\tau} - T_{jt+\tau}] \\ & \left. + \lambda_{t+\tau}^k \left[\varepsilon_{t+\tau}^I I_{jt+\tau} \left(1 - f \left(\frac{I_{jt+\tau}}{I_{jt-1+\tau}} \right) \right) + (1 - \delta) K_{jt-1+\tau} - K_{jt+\tau} \right] \right\} \\ - & \end{aligned}$$

If you're not comfortable with infinite sums, you can develop the sum for $\tau = 0$ and $\tau = 1$ (we don't mind for value of tau above 1):

$$\begin{aligned}
\mathcal{L}_t = & \varepsilon_t^C \left(\frac{C_{jt}^{1-\sigma_C}}{1-\sigma_C} - \chi \frac{H_{jt}^{1+\sigma_L}}{1+\sigma_L} \right) \\
& + \varepsilon_t^C \lambda_t^c [W_t H_{jt} + Z_t K_{jt-1} + R_{t-1} B_{jt-1} - C_{jt} - I_{jt} - B_{jt} - T_{jt}] \\
& + \varepsilon_t^C \lambda_t^k \left[\varepsilon_t^I I_{jt} \left(1 - f \left(\frac{I_{jt}}{I_{jt-1}} \right) \right) + (1-\delta) K_{jt-1} - K_{jt} \right] \\
& + \varepsilon_{t+1}^C \beta \left(\frac{C_{jt+1}^{1-\sigma_C}}{1-\sigma_C} - \chi \frac{H_{jt+1}^{1+\sigma_L}}{1+\sigma_L} \right) \\
& + \varepsilon_{t+1}^C \lambda_{t+1}^c \beta [W_{t+1} H_{jt+1} + Z_{t+1} K_{jt} + R_t B_{jt} - C_{jt+1} - I_{jt+1} - B_{jt+1} - T_{jt+1}] \\
& + \varepsilon_{t+1}^C \lambda_{t+1}^k \beta \left[\varepsilon_{t+1}^I I_{jt+1} \left(1 - f \left(\frac{I_{jt+1}}{I_{jt}} \right) \right) + (1-\delta) K_{jt} - K_{jt+1} \right] \\
& + \varepsilon_{t+2}^C \beta^2 \left(\frac{C_{jt+2}^{1-\sigma_C}}{1-\sigma_C} - \chi \frac{H_{jt+2}^{1+\sigma_L}}{1+\sigma_L} \right) \\
& + \dots
\end{aligned}$$

Optimal condition on investment:

$$-\varepsilon_t^C \lambda_t^c + \varepsilon_t^C \lambda_t^k \varepsilon_t^I \frac{\partial [I_{jt} (1 - AC_{jt}^I)]}{\partial I_{jt}} - \varepsilon_{t+1}^C \lambda_{t+1}^k \beta \varepsilon_{t+1}^I \frac{\partial [I_{jt+1} AC_{jt+1}^I]}{\partial I_{jt}} = 0$$

Only one first order conditions is different from what we had in the standard RBC model:

$$\begin{aligned}
(\partial C_{jt}) : \lambda_t^c &= C_{jt}^{-\sigma_C} \\
(\partial H_{jt}) : \chi H_{jt}^{\sigma_L} &= \lambda_t^c W_t \\
(\partial B_{jt}) : \mathbb{E}_t \left\{ \frac{\varepsilon_t^C \lambda_t^c}{\varepsilon_{t+1}^C \lambda_{t+1}^c} \right\} &= \beta R_t \\
(\partial I_{jt}) : \frac{\lambda_t^k}{\lambda_t^c} \varepsilon_t^I \frac{\partial [I_{jt} (1 - AC_{jt}^I)]}{\partial I_{jt}} - \beta \frac{\varepsilon_{t+1}^C \lambda_{t+1}^k}{\varepsilon_t^C \lambda_t^c} \varepsilon_{t+1}^I \frac{\partial [I_{jt+1} AC_{jt+1}^I]}{\partial I_{jt}} &= 1 \\
(\partial K_{jt}) : \varepsilon_t^C \lambda_t^k &= \beta \mathbb{E}_t \left\{ \varepsilon_{t+1}^C \left[(1-\delta) \lambda_{t+1}^k + \lambda_{t+1}^c Z_{t+1} \right] \right\}
\end{aligned}$$

We define the asset price q_t by the relative ratio of the marginal utility of buying capital goods λ_t^k over the marginal utility of consumption λ_t^c with $q_t = \lambda_t^k / \lambda_t^c$. When $q_t > 1$, households are willing to invest in physical capital rather than consuming, this can be interpreted as an increase in asset prices with respect to its steady state. We can thus replace λ_t^k by the asset price equation $\lambda_t^k = q_t \lambda_t^c$. The substitution affects first order conditions on (∂I_{jt}) and (∂K_{jt}) :

$$\begin{aligned}
(\partial I_{jt}) : \frac{\lambda_t^k}{\lambda_t^c} \varepsilon_t^I \frac{\partial [I_{jt} (1 - AC_{jt}^I)]}{\partial I_{jt}} - \beta \frac{\varepsilon_{t+1}^C \lambda_{t+1}^k}{\varepsilon_t^C \lambda_t^c} \varepsilon_{t+1}^I \frac{\partial [I_{jt+1} AC_{jt+1}^I]}{\partial I_{jt}} &= 1 \\
: \frac{q_t \lambda_t^c}{\lambda_t^c} \varepsilon_t^I \frac{\partial [I_{jt} (1 - AC_{jt}^I)]}{\partial I_{jt}} - \beta \frac{\varepsilon_{t+1}^C q_{t+1} \lambda_{t+1}^c}{\varepsilon_t^C \lambda_t^c} \varepsilon_{t+1}^I \frac{\partial [I_{jt+1} AC_{jt+1}^I]}{\partial I_{jt}} &= 1 \\
: q_t \varepsilon_t^I \frac{\partial [I_{jt} (1 - AC_{jt}^I)]}{\partial I_{jt}} - \frac{1}{R_t} \mathbb{E}_t q_{t+1} \varepsilon_{t+1}^I \frac{\partial [I_{jt+1} AC_{jt+1}^I]}{\partial I_{jt}} &= 1
\end{aligned}$$

and for (∂K_{jt}) :

$$\begin{aligned}
(\partial K_{jt}) : \varepsilon_t^C q_t \lambda_t^c &= \beta \mathbb{E}_t \left\{ \varepsilon_{t+1}^C \left[(1-\delta) q_{t+1} \lambda_{t+1}^c + \lambda_{t+1}^c Z_{t+1} \right] \right\} \\
: q_t &= \beta \mathbb{E}_t \left\{ \frac{\lambda_{t+1}^c \varepsilon_{t+1}^C}{\varepsilon_t^C \lambda_t^c} \left[(1-\delta) q_{t+1} + Z_{t+1} \right] \right\} \\
: q_t &= \mathbb{E}_t \left\{ \frac{1}{R_t} \left[(1-\delta) q_{t+1} + Z_{t+1} \right] \right\} \\
: R_t &= \frac{(1-\delta) \mathbb{E}_t q_{t+1} + \mathbb{E}_t Z_{t+1}}{q_t}
\end{aligned}$$

these two equations are really popular in DSGE modelling. Many papers re-employ this investment cost function as it generate a dynamic of investment consistent with empirical evidence and data friendly (response of invesment is hump-shaped as in VARs models).

2 Model summary

We have to include one new variables $\{q_t\}$, and one corresponding equation to obtain a determinate system of equations:

$$\begin{aligned}
& : \beta R_t = \mathbb{E}_t \left\{ \frac{\varepsilon_t^C}{\varepsilon_{t+1}^C} \left(\frac{C_{jt+1}}{C_{jt}} \right)^{\sigma_C} \right\} \\
& : W_t = \chi H_t^{\sigma_L} C_t^{\sigma_C} \\
& : R_t = \frac{\mathbb{E}_t \{ (1 - \delta) q_{t+1} + Z_{t+1} \}}{q_t} \\
& : K_t = \varepsilon_t^I \left(1 - f \left(\frac{I_{jt}}{I_{jt-1}} \right) \right) I_t + (1 - \delta) K_{t-1} \\
& : q_t \varepsilon_t^I \frac{\partial [I_{jt} (1 - AC_{jt}^I)]}{\partial I_{jt}} - \frac{1}{R_t} \mathbb{E}_t q_{t+1} \varepsilon_{t+1}^I \frac{\partial [I_{jt+1} AC_{jt+1}^I]}{\partial I_{jt}} = 1 \\
& : Y_t = \varepsilon_t^A K_{t-1}^\alpha H_t^{1-\alpha}, \\
& : Z_t = \alpha \frac{Y_t}{K_{t-1}} \\
& : W_t = (1 - \alpha) \frac{Y_t}{H_t} \\
& : Y_t = C_t + I_t + g^y \bar{Y} \varepsilon_t^G \\
& : \log(\varepsilon_t^A) = \rho_A \log(\varepsilon_{t-1}^A) + \eta_t^A \\
& : \log(\varepsilon_t^G) = \rho_G \log(\varepsilon_{t-1}^G) + \eta_t^G \\
& : \log(\varepsilon_t^C) = \rho_C \log(\varepsilon_{t-1}^C) + \eta_t^C \\
& : \log(\varepsilon_t^I) = \rho_I \log(\varepsilon_{t-1}^I) + \eta_t^I
\end{aligned}$$

3 Steady state

The asset price equation is normalized to unity in steady state:

$$\bar{q} = 1$$

4 Dynare writing

1. Declare one new endogenous variable $\{q_t\}$
2. Declare the investment cost χ as a parameter and calibrate it to $\chi = 4$
3. Include the one dynamic equation defining $\{q_t\}$ and modify equations affected by capital utilization.
4. Define the steady state of $\{q_t\}$ in the init block.

5 Tips

First, regarding investment costs derivation:

$$\begin{aligned}
\frac{\partial [I_t AC_t^I]}{\partial I_t} &= \frac{\chi}{2} \left(3 \left(\frac{I_t}{I_{t-1}} \right)^2 + 1 - 4 \frac{I_t}{I_{t-1}} \right), \\
\frac{\partial [I_{t+1} AC_{t+1}^I]}{\partial I_t} &= \chi \left(\left(\frac{I_{t+1}}{I_t} \right)^2 - \left(\frac{I_{t+1}}{I_{t+1}} \right)^3 \right).
\end{aligned}$$

Second you may have an headache reading this equation:

$$: q_t \varepsilon_t^I \frac{\partial [I_{jt} (1 - AC_{jt}^I)]}{\partial I_{jt}} - \frac{1}{R_t} \mathbb{E}_t q_{t+1} \varepsilon_{t+1}^I \frac{\partial [I_{jt+1} AC_{jt+1}^I]}{\partial I_{jt}} = 1$$

however, up to first order, this equation boils down to:

$$\hat{q}_t = \chi(\hat{i}_t - \hat{i}_{t-1}) - \beta \chi(\hat{i}_{t+1} - \hat{i}_t)$$